

Phạm Thanh Lâm

Network Engineer · Computer Networks & Data Communications

Network Engineer focused on designing and validating complete solutions across network, security, and cloud environments. Experienced in building real-world systems end-to-end. Seeking Network/Security Engineering roles to contribute to practical system design and operations.

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EXPERIENCE

FSI Technology Joint Stock Company

Mar 2026 - Jun 2026

Network & Security Engineer Intern

26/22 Street No. 03, Do Thanh Residential Area, Ban Co Ward, Ho Chi Minh City

- Deployed and validated F-DDH BOX security appliance across lab and production environments — traffic simulation, alert verification, firmware coordination, and presales support.
- Researched VPN evasion techniques (Tailscale, Cloudflare WARP, WireGuard) to assess NGFW detection gaps; produced comparative analysis for enterprise security advisory.
- Supported Cisco network initiatives: switch upgrade research, StackWise stacking, and SD-WAN evaluation on Cisco dCloud; assisted with on-site infrastructure deployment and fiber troubleshooting.
- Deployed internal attendance web app on local server — NAT port forwarding, custom domain setup, functional testing, and basic security assessment.

PROJECTS

HA Firewall & AI-driven Security Monitoring — Graduation Thesis

Jan - May 2026

- Deployed Active-Passive OPNsense cluster (CARP/pfsync) with HAProxy SSL offloading and ModSecurity WAF (OWASP CRS); 2.8s failover (<5s target), zero-session-drop with pfsync, and only 0.71% sync overhead.
- Executed WAF validation with Nikto scan: 12,796 alerts, DR 100% / FPR 0% on the baseline payload set; conducted payload-level analysis to quantify security vs. false-positive trade-offs.
- Built a Wazuh SIEM pipeline and trained Random Forest / Logistic Regression on a custom attack dataset (DVWA + Kali Linux): average detection time 1.80s, RF AUC 1.000 (lab), LR AUC 0.981.

OPNsense CARP/pfsync HAProxy ModSecurity OWASP CRS Wazuh SIEM Random Forest Python VMware

Cisco SD-WAN Hospital Network — Cisco dCloud Lab

Apr 2026

- Architected a 3-site Hub-and-Spoke SD-WAN overlay on Cisco dCloud CML (2x C8000v); enforced IPsec AES-256-GCM across both transports (MPLS + biz-internet) — guaranteeing patient data is never transmitted in plaintext, even during transport failover.
- Implemented least-privilege traffic segmentation via AAR policy: pinned critical medical traffic (Oracle, MS-SQL) exclusively to MPLS, fully isolated from the internet-bound path; verified failover exposes no sensitive traffic through vManage Events Log — ~3 min convergence, zero manual intervention.

Cisco CML Catalyst SD-WAN 20.18 vManage C8000v AAR Policy IPsec BFD Hub-and-Spoke

AI Shield Firewall — LLM Traffic Inspection Proxy — Personal Project

2025 - 2026

- Built a 5-layer bilingual (EN+VI) LLM security proxy on AWS (FastAPI, Docker) enforcing DLP at enterprise AI API perimeters: Regex → Presidio NLP + Vietnamese PII recognizers (CCCD, tax code, license plate) → Prompt Injection detection → per-org Policy Engine → Output Inspection; 25/25 unit tests passed, <15ms proxy overhead, IAM Role credentials + DynamoDB audit logging.

Python FastAPI AWS EC2 Docker Presidio NLP Prompt Injection IAM Role DynamoDB

AutoCTI — Autonomous Cyber Threat Intelligence Agent — Personal Project

2025 - 2026

- Built a 24/7 automated SOC threat-intelligence pipeline on AWS Free Tier (~\$0/month): LangGraph ReAct Agent + Gemini 2.5 Flash + Tavily real-time search for autonomous threat triage and Telegram alerting; IAM Role credentials, DynamoDB conversation memory, and production-grade reliability achieving ~80% RAM reduction over prior Playwright-based architecture.

AWS SQS LangGraph Python Docker Gemini AI Telegram Bot API

CERTIFICATIONS

Networking Devices & Initial Configuration

Cisco NetAcad · 2026

Linux Essentials

Cisco NetAcad · 2026

EDUCATION

University of Information Technology (UIT – VNU–HCM)

B.Sc. in Computer Networks and Data Communications

GPA 3.09 / 4.0 · 2022 - 2026

SKILLS

NETWORKING

- TCP/IP
- Routing/Switching
- SD-WAN
- High Availability
- Cisco IOS
- OSPF
- BGP
- VLAN/STP
- ACL/NAT
- BFD
- CARP/pfsync
- IPv6

SECURITY

- OPNsense
- Wazuh SIEM
- AI Security
- ModSecurity WAF
- Suricata IDS/IPS
- HAProxy
- Nmap
- Zabbix

CLOUD & INFRA

- AWS
- Docker
- Linux
- VMware
- Cisco CML

DEV & TOOLS

- Python
- C++
- FastAPI
- Git
- REST API
- Shell Script
- Wireshark

LANGUAGES

Vietnamese Native

English Technical reading